

Agentic-AI

Code Execution

Gefördert durch:



Bundesministerium
für Forschung, Technologie
und Raumfahrt



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Example

For example, the model might generate:

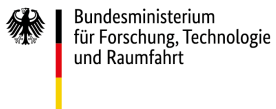
```
python  
  
nums = [10, 15, 25]  
print(sum(nums) / len(nums))
```

The sandbox executes it and returns:

```
16.6666
```

The model then responds with the precise value, not an approximation.

Gefördert durch:



- **Augments its own capabilities**
Can run code even when unable to do a task directly
- **Generates functions on demand**
No need to predefine every function for the LLM to use

Code Executor Tool

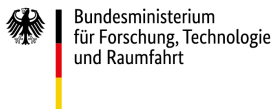
```
def execute_code(code: str):
    try:
        # Capture variables created by the code
        local_vars = {}
        exec(code, {}, local_vars)

        return {
            "output": local_vars.get("result", None),
            "success": True
        }

    except Exception as e:
        return {
            "output": str(e),
            "success": False
        }
```

```
code = "result = (10 + 15 + 25) / 3"
execute_code(code)
# → {'output': 16.6666667, 'success': True}
```

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Sandboxed Execution

Code Snippet 1 — Separate Process

python

 Copy code

```
subprocess.Popen(["python3", "-c", code], ...)
```

Code Snippet 2 — Container

bash

 Copy code

```
docker run --rm --network=none -v /empty_dir:/app python:3.11 python -c "$code"
```

Code Snippet 3 — Sandboxed Service

python

 Copy code

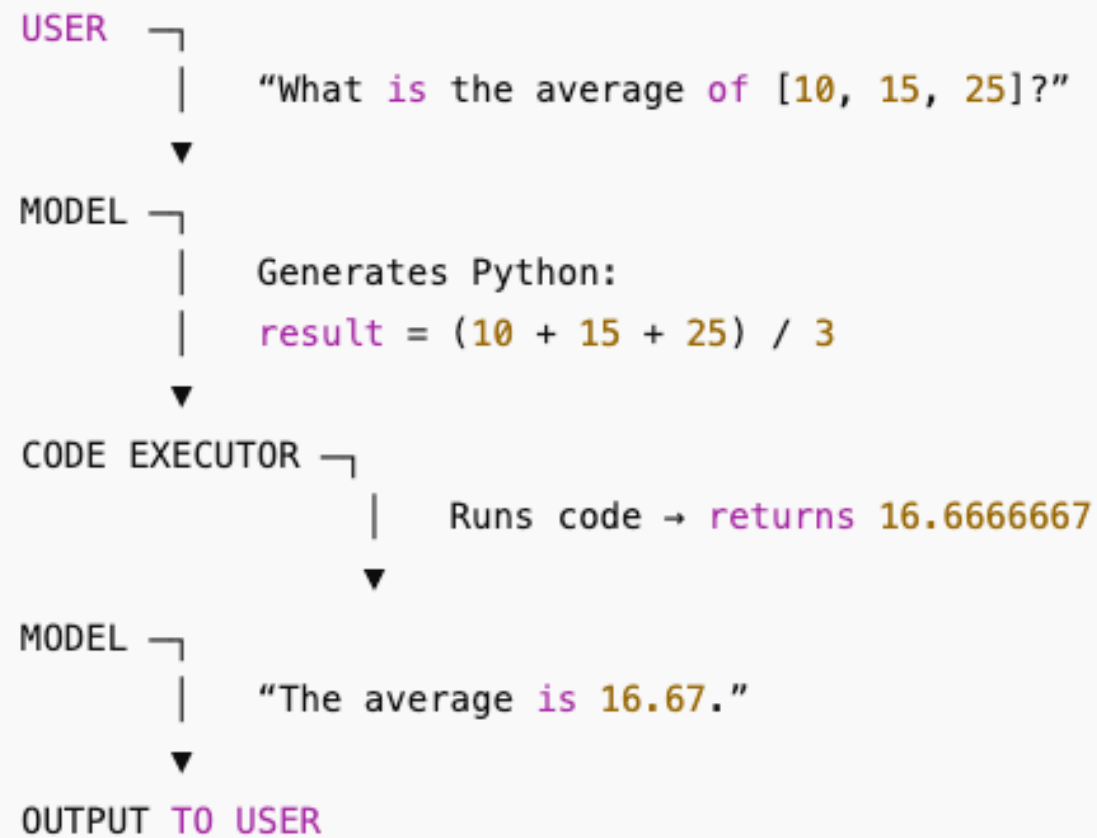
```
run_sandboxed(code)
```

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End-to-End Example Overview



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